

A PROJECT TITLE
ON
Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
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by

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1. Abstract:

Customer churn, the phenomenon of subscribers or clients discontinuing use of a company's product or service, poses a significant threat to revenue and long-term growth in subscription-driven and service-oriented industries such as telecommunications, banking, and e-commerce. Early and accurate identification of customers who are likely to churn enables organizations to design targeted retention strategies and reduce acquisition costs. This project proposes a churn prediction framework built on Elastic Net Regularized Logistic Regression, a technique that combines L1 (Lasso) and L2 (Ridge) penalties within the logistic regression objective function. The Elastic Net penalty addresses the limitations of using L1 or L2 regularization in isolation: it performs automatic feature selection by shrinking irrelevant coefficients to zero while simultaneously handling multicollinearity among correlated predictors, which is common in customer behavioral datasets containing usage patterns, billing history, tenure, and service subscriptions. The dataset is preprocessed through handling of missing values, encoding of categorical variables, and standardization of numerical features before being split into training and testing subsets. The model is trained using cross-validation to tune the regularization strength (λ) and the mixing parameter (α) that balances the L1 and L2 contributions, optimizing for metrics such as accuracy, precision, recall, F1-score, and the area under the ROC curve. Comparative experiments against plain logistic regression, decision trees, and random forest classifiers demonstrate that the elastic net approach yields a more interpretable and generalizable model without significant loss in predictive performance, while also reducing overfitting on high-dimensional customer data. The resulting model identifies the most influential factors driving churn, such as contract type, monthly charges, and customer tenure, providing actionable insights for business stakeholders. The proposed system can be integrated into a customer relationship management pipeline to flag at-risk customers in real time, allowing proactive retention interventions. Overall, this work demonstrates that Elastic Net Regularized Logistic Regression offers a robust, interpretable, and computationally efficient solution for customer churn prediction in real-world business environments.

2. INTRODUCTION:

Customer retention has become a critical priority for businesses operating in highly competitive, subscription-based markets. Acquiring a new customer is widely recognized to be considerably more expensive than retaining an existing one, which makes the ability to predict churn, the likelihood that a customer will discontinue a service, a valuable capability for any organization. Traditional approaches to churn analysis relied on manual rule-based systems or simple statistical summaries, which often failed to capture complex, non-linear relationships hidden within large volumes of customer data. With the growth of machine learning, predictive models have become the preferred approach for churn analysis due to their ability to learn patterns from historical customer behavior and generalize to new,

unseen customers. Logistic regression remains a popular choice for churn prediction because of its interpretability and probabilistic output, which allows business analysts to understand not just who is likely to churn, but why. However, real-world customer datasets often contain a large number of correlated features, such as multiple billing and usage-related attributes, which can destabilize a standard logistic regression model and lead to overfitting. Elastic Net regularization addresses this challenge by combining the L1 penalty, which promotes sparsity and automatic feature selection, with the L2 penalty, which stabilizes coefficient estimates in the presence of correlated predictors. This hybrid approach makes it particularly well-suited to churn datasets that mix demographic, transactional, and service-usage features. This project explores the design, implementation, and evaluation of an Elastic Net Regularized Logistic Regression model for predicting customer churn, with the goal of building a model that is both accurate and interpretable enough to support data-driven retention strategies.

3. SOFTWARE REQUIREMENTS:

- Programming Language: Python 3.10 or above
- Libraries: scikit-learn, pandas, NumPy, Matplotlib, Seaborn
- Development Environment: Jupyter Notebook / Google Colab / VS Code
- Version Control: Git and GitHub
- Operating System: Windows 10/11, Linux, or macOS
- Dataset Name: Telco Customer Churn Dataset
- Dataset Format: CSV (Comma Separated Values)

Signature of the Guide

Course Instructor